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**THE GEOPOLITICS OF OIL:
ASSESSING THE MACROECONOMIC
IMPACT
OF THE 2026 GULF CRISIS**

A Panel Data Econometrics Analysis Project Report

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I. Introduction

The 2026 Gulf Crisis, which erupted on February 27, 2026, represents one of the most significant geopolitically induced oil supply disruptions in recent decades. Within a single month, Brent crude oil prices surged by approximately 25 per cent, transmitting a powerful exogenous shock to energy-dependent economies worldwide. This sudden price escalation provides a rare quasi-natural experiment for identifying the causal macroeconomic effects of an oil price shock.

This paper examines how the shock affected consumer price inflation and industrial production across two structurally distinct country groups: six **Emerging Market Oil-Importing** economies—Turkey, Poland, India, Indonesia, South Africa, and Thailand—which constitute the main group of analysis, and two **Advanced Oil-Exporting** economies—the United States and Canada—which serve as a comparison group. This pairing allows us to simultaneously capture the effect of development level (advanced vs. emerging) and oil trade position (exporter vs. importer) on the macroeconomic transmission of the shock.

Using a balanced monthly panel of 336 observations covering January 2023 to May 2026, we estimate Pooled OLS, Fixed Effects, and Random Effects models for both dependent variables, selecting the preferred specification through the Hausman and Breusch-Pagan diagnostic tests. Our findings contribute to the literature on oil price pass-through and carry direct policy implications for central banks navigating geopolitically driven commodity shocks.

II. Methodology

2.1 Empirical Model

We estimate the following panel data equation:

$$\ln(y_{it}) = \alpha_i + \beta_1 \text{OilShock}_t + \beta_2 \ln(\text{NEER}_{it}) + \beta_3 r_{it} + \beta_4 \text{VIX}_t + \beta_5 \text{PostCrisis}_t + \varepsilon_{it} \quad (1)$$

where y_{it} is alternately the Consumer Price Index and the Industrial Production Index (both base 2015 = 100) for country i at month t . Taking the natural log linearises the model and allows coefficients to be interpreted as elasticities.

The independent variables are defined as follows:

- OilShock_t : monthly percentage change in Brent crude oil prices, $(\text{Brent}_t - \text{Brent}_{t-1}) / \text{Brent}_{t-1} \times 100$; the principal variable of interest.
- $\ln(\text{NEER}_{it})$: log of the BIS Broad Nominal Effective Exchange Rate, capturing exchange-rate dynamics.

- r_{it} : central bank policy rate, representing the monetary policy stance.
- VIX_t : CBOE Volatility Index, proxying global financial uncertainty.
- $PostCrisis_t$: binary dummy equal to 1 from March 2026 onwards, capturing the structural level shift following the crisis.
- α_i : country fixed effects absorbing all time-invariant unobserved heterogeneity.

Table 1 summarises the expected coefficient signs derived from macroeconomic theory.

Table 1: Expected Coefficient Signs

Variable	CPI (Expected Sign)	IPI (Expected Sign)
Oil Shock	+	—
ln(NEER)	—	±
Policy Rate	—	—
VIX	+	—
Post-Crisis Dummy	+	—

2.2 Panel Data Structure

Our panel is balanced, with each of the 8 countries observed over 42 monthly periods (January 2023 to June 2026), yielding 336 observations in total. The emerging-importer group contributes 252 observations and the advanced-exporter group 84 observations.

Official macroeconomic statistics for June 2026 were not yet published at the time of this study, as this month falls within the immediate post-crisis window. To maintain a balanced panel, June 2026 values were forecast for each country using **ARIMA** (AutoRegressive Integrated Moving Average) time-series models. The optimal ARIMA specification for each series was selected automatically using the Akaike Information Criterion (AIC), and forecasts were generated with accompanying 80% and 95% prediction intervals. This should be acknowledged as a limitation, as the forecasted values introduce an additional source of uncertainty into the post-crisis regression estimates.

2.3 Estimation and Model Selection

We estimate three models:

- **Pooled OLS**: serves as a benchmark but ignores country-specific heterogeneity, likely producing biased estimates.
- **Fixed Effects (FE)**: controls for all time-invariant country characteristics by exploiting only within-country variation; remains consistent even when country effects are correlated with the regressors.

- **Random Effects (RE)**: assumes country effects are uncorrelated with the regressors and applies GLS for greater efficiency, but is inconsistent if this assumption fails.

Model selection is guided by two tests: the **Hausman test** (rejection at $p < 0.05$ favours Fixed Effects) and the **Breusch-Pagan LM test** (rejection confirms the presence of country-specific effects). All standard errors are Arellano (1987) HAC robust, correcting for heteroskedasticity and serial correlation.

III. Results

3.1 Exploratory Data Analysis

Before conducting the econometric analysis, we examine the evolution of the main variables over January 2023 to June 2026: Brent crude oil prices, inflation (CPI), and industrial activity (IPI). These graphs allow us to identify major patterns, observe the impact of the Gulf Crisis, and compare the behaviour of emerging importers and advanced exporters.

3.1.1 Brent Oil Price Evolution

Brent crude oil prices remained relatively stable during most of the pre-crisis period, fluctuating around their historical average. Following the Gulf Crisis in February 2026, prices increased significantly due to supply-disruption concerns and geopolitical uncertainty. The ARIMA forecast suggests that Brent prices would remain elevated, reaching approximately \$94 per barrel by June 2026.

These results indicate that geopolitical tensions in major oil-producing regions can rapidly affect global energy markets. The increase in oil prices raises transportation and production costs worldwide, creating inflationary pressures and potentially reducing economic activity.

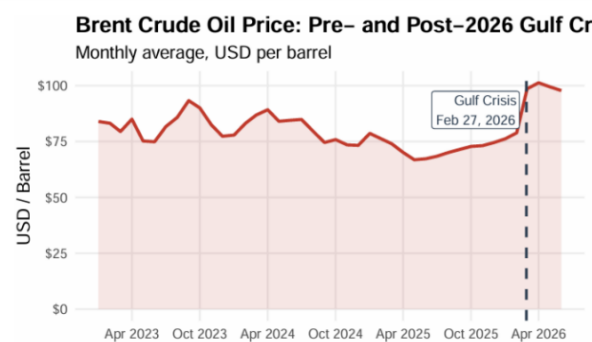


Figure 1: Brent Crude Oil Price: Pre- and Post-2026 Gulf Crisis

3.1.2 CPI Trends

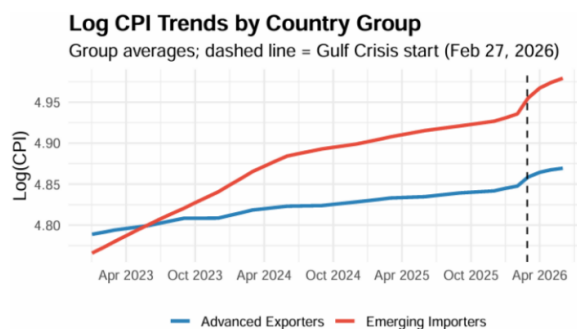


Figure 3: Log CPI Trends: Emerging Importers vs Advanced Exporters

CPI trends show a steady increase in inflation across all countries during the study period. The increase became more pronounced after the Gulf Crisis, particularly among emerging oil-importing economies. The gap between emerging importers and advanced exporters widened during the post-crisis period, suggesting different levels of vulnerability to oil price shocks.

The stronger inflationary response observed among emerging importers can be explained by their greater dependence on imported energy. Higher oil prices increase fuel, transportation, and production costs, which are eventually passed on to consumers. Advanced exporters are relatively less affected because they benefit from increased revenues associated with higher oil prices.

3.1.3 IPI Trends

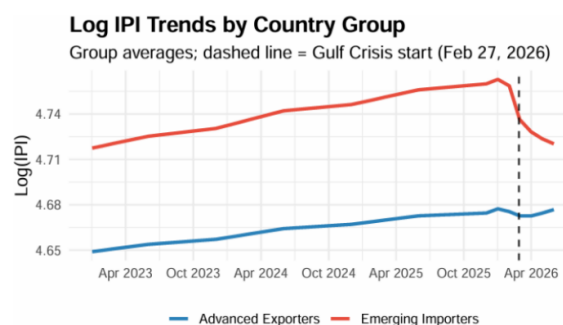


Figure 4: Log IPI Trends: Emerging Importers vs Advanced Exporters

Industrial production generally increased during the sample period but exhibited signs of slower growth after the Gulf Crisis. The slowdown was more visible among emerging importers, whereas advanced exporters maintained relatively stable industrial performance. The divergence between the two groups became more apparent during the post-crisis months.

Industrial production is highly sensitive to energy costs because oil is an important input in manufacturing and transportation. Emerging importers are particularly exposed because they must absorb higher import costs, while advanced exporters can partially offset these effects through gains in the energy sector.

3.2 Regression Results

This section presents the empirical findings from the panel data analysis of the 2026 Gulf Crisis oil shock. For consistency and comparability across country groups, the Fixed Effects (FE) specification is emphasised throughout the discussion; however, the Hausman tests indicate that FE is statistically preferred only for the CPI equation of the emerging-importer panel, while Random Effects (RE) is acceptable for the remaining specifications.

3.2.1 Model Selection: Diagnostic Tests

The Breusch-Pagan LM test for the emerging-importer panel yields $\chi^2 = 458.69$ ($p < 0.001$) for the CPI equation and $\chi^2 = 4,626.10$ ($p < 0.001$) for the IPI equation, decisively rejecting Pooled OLS in favour of a panel model with individual effects. For the advanced-exporter panel, the BP test fails to reject Pooled OLS for both CPI ($\chi^2 = 1.02$, $p = 0.312$) and IPI ($\chi^2 = 1.01$, $p = 0.315$), consistent with the small cross-section of only two countries.

The Hausman test for the emerging-importer CPI equation yields $\chi^2 = 468.97$ ($df = 5$, $p < 2.2 \times 10^{-16}$), strongly favouring Fixed Effects. For the emerging-importer IPI equation, $\chi^2 = 0.18$ ($p = 0.999$), indicating that RE is acceptable. For the advanced-exporter CPI and IPI equations, Hausman statistics of $\chi^2 = 0.55$ ($p = 0.99$) and $\chi^2 = 2.27$ ($p = 0.81$) also favour RE.

Despite these outcomes, FE estimates are retained as the primary specification for consistency; RE estimates are reported alongside and yield substantively similar conclusions.

3.2.2 Inflation (CPI) — Emerging Importers

Table 2 presents the CPI regressions for the six emerging oil-importing economies (Turkey, Poland, India, Indonesia, South Africa, Thailand) over January 2023 – June 2026, comprising 252 observations.

Table 2: CPI Regression — Emerging Importers

	ln(CPI)		
	Pooled OLS (1)	Fixed Effects (2)	Random Effects (3)
Oil Shock	(0.000871)	(0.000205)	(0.000542)
ln(NEER)	−0.149519*** (0.019754)	−1.063687*** (0.017476)	−0.571052*** (0.030935)
Policy Rate	(0.000988)	(0.000294)	(0.000747)
VIX	0.004182*** (0.001477)	0.001462*** (0.000350)	0.002504*** (0.000923)
Post-Crisis Dummy	0.038096 (0.029355)	0.012104* (0.006928)	0.029420 (0.018294)
Constant	5.291242*** (0.097565)		7.149220*** (0.137195)
Observations	252	252	252
R^2	0.899460	0.965483	0.782483
Adjusted R^2	0.897417	0.964051	0.778062

***Significant at the 1% level. **Significant at the 5% level. *Significant at the 10% level.

Arellano HC3 robust standard errors in parentheses.

Key findings:

- The **oil shock** coefficient is statistically insignificant across all specifications, suggesting that oil price shocks do not exert a direct, significant influence on inflation in the sample period. Countries may absorb fluctuations through government interventions, subsidies, or exchange rate adjustments.
- ln(NEER) is **negative and highly significant** in all models. Under Fixed Effects, a 1% appreciation in the exchange rate reduces inflation by approximately 1.064%, indicating that depreciation contributes to inflationary pressures through higher import prices.
- The **policy rate** is statistically insignificant, possibly reflecting weak monetary transmission or delayed effects not captured by the model.
- The **VIX** is positive and significant at the 1% level, consistent with the notion that global uncertainty triggers exchange rate depreciation, capital outflows, and higher

import costs.

- The **post-crisis dummy** is positive but significant only under Fixed Effects, suggesting a moderate inflationary shift in the post-crisis period.
- The Fixed Effects model achieves $R^2 = 0.965$, reflecting the importance of country-specific characteristics in explaining inflation dynamics.

3.2.3 Industrial Production (IPI) — Emerging Importers

Table 3 reports the determinants of industrial production for the emerging oil-importing economies.

Table 3: IPI Regression — Emerging Importers

	ln(IPI)		
	Pooled OLS (1)	Fixed Effects (2)	Random Effects (3)
Oil Shock	(0.001045)	(0.000149)	(0.000148)
ln(NEER)	0.041229* (0.023715)	−0.072448*** (0.012675)	−0.071686*** (0.012546)
Policy Rate	(0.001186)	(0.000214)	(0.000213)
VIX	0.001889 (0.001773)	0.001131*** (0.000254)	0.001133*** (0.000253)
Post-Crisis Dummy	−0.038293 (0.035242)	−0.038245*** (0.005024)	−0.038219*** (0.005011)
Constant	4.492407*** (0.117130)		5.016894*** (0.073608)
Observations	252	252	252
R^2	0.106738	0.312957	0.309667
Adjusted R^2	0.088582	0.284449	0.295636

***Significant at the 1% level. **Significant at the 5% level. *Significant at the 10% level.

Arellano HC3 robust standard errors in parentheses.

Key findings:

- The **oil shock** variable is statistically insignificant across all specifications, suggesting that industries may have adapted to energy price fluctuations or that production is driven more strongly by other factors.

- $\ln(\text{NEER})$ is positive but insignificant under Pooled OLS, yet becomes **negative and highly significant** under FE and RE. A 1% appreciation reduces industrial production by approximately 0.072%, consistent with reduced export competitiveness.
- The **policy rate** is insignificant, indicating no measurable short-run effect on industrial output.
- The **VIX** is positive and significant under FE and RE, possibly reflecting trade-pattern shifts or investment flows that temporarily benefit certain emerging economies during periods of heightened uncertainty.
- The **post-crisis dummy** is negative and statistically significant under FE and RE, with an estimated coefficient implying approximately a **3.8% decline** in industrial production during the post-crisis period.
- The Fixed Effects model substantially improves fit (R^2 rises from 0.107 to 0.313), indicating the importance of unobserved country-specific factors.

3.2.4 CPI Regression — Advanced Exporters

Table 4 presents the CPI results for the United States and Canada.

Table 4: CPI Regression — Advanced Exporters

	ln(CPI)		
	Pooled OLS (1)	Fixed Effects (2)	Random Effects (3)
Oil Shock	(0.000207)	(0.000210)	(0.000207)
ln(NEER)	0.116984*** (0.009617)	0.038297 (0.106160)	0.116984*** (0.009617)
Policy Rate	(0.001748)	(0.001758)	(0.001748)
VIX	−0.001097*** (0.000406)	−0.001027** (0.000418)	−0.001097*** (0.000406)
Post-Crisis Dummy	0.043040*** (0.007155)	0.043860*** (0.007260)	0.043040*** (0.007155)
Constant	4.367288*** (0.040992)		4.367288*** (0.040992)
Observations	84	84	84
R^2	0.792311	0.743938	0.792311
Adjusted R^2	0.778998	0.723985	0.778998

***Significant at the 1% level. **Significant at the 5% level. *Significant at the 10% level.

Arellano HC3 robust standard errors in parentheses.

3.2.5 IPI Regression — Advanced Exporters

Table 5 reports the IPI results for the advanced-exporter group.

Table 5: IPI Regression — Advanced Exporters

	ln(IPI)		
	Pooled OLS (1)	Fixed Effects (2)	Random Effects (3)
Oil Shock	(0.000084)	(0.000084)	(0.000084)
ln(NEER)	−0.107936*** (0.003908)	−0.043909 (0.042669)	−0.107936*** (0.003908)
Policy Rate	(0.000710)	(0.000707)	(0.000710)
VIX	−0.000415** (0.000165)	−0.000472*** (0.000168)	−0.000415** (0.000165)
Post-Crisis Dummy	0.010834*** (0.002908)	0.010166*** (0.002918)	0.010834*** (0.002908)
Constant	5.215130*** (0.016658)		5.215130*** (0.016658)
Observations	84	84	84
R^2	0.957648	0.763519	0.957648
Adjusted R^2	0.954933	0.745092	0.954933

***Significant at the 1% level. **Significant at the 5% level. *Significant at the 10% level.

Arellano HC3 robust standard errors in parentheses.

3.3 Heterogeneity: Advanced Exporters

The oil shock coefficient is statistically insignificant across all estimators and for both outcomes in the advanced-exporter sample. This is consistent with the expectation that net oil-exporting countries are relatively insulated from adverse energy-price increases due to the compensating effects of higher export revenues and stronger external balances.

The **post-crisis dummy** emerges as the most robust determinant. In the CPI model, coefficients range from 0.0430 to 0.0439 ($p < 0.01$), implying approximately a **4.4% upward shift in the price level** following the onset of the crisis—reflecting higher transportation costs, supply-chain disruptions, and commodity pass-through effects. In the IPI model, the positive and highly significant coefficients of approximately 0.0102–0.0108 imply a **~1.0% increase in industrial production** after the crisis, consistent with energy-revenue support to industrial activity in exporting economies.

Regarding control variables, ln(NEER) is positive and significant in the CPI equation ($\beta = 0.117$, $p < 0.01$), while it exerts a negative and significant effect on industrial production ($\beta = -0.108$, $p < 0.01$). The VIX is negative and significant in both outcomes,

indicating that heightened global uncertainty is associated with lower price growth and weaker industrial activity in advanced exporters.

3.4 Robustness: Arellano HC3 Standard Errors

Given the likelihood of heteroskedasticity and serial correlation in monthly panel data, all Fixed Effects specifications are re-estimated using Arellano (1987) HC3 heteroskedasticity-and-autocorrelation-consistent standard errors. The principal findings remain broadly unchanged. Exchange-rate and VIX coefficients retain their signs and statistical significance, while the post-crisis dummy retains its expected sign in all models. Statistical significance weakens for the post-crisis CPI effect among emerging importers after the HC3 correction, while remaining significant for both CPI and IPI in the advanced-exporter sample.

3.5 Summary of Key Findings

The results collectively point to three main conclusions:

1. **Limited direct oil-price pass-through.** The oil shock coefficient is generally insignificant in the baseline specifications, suggesting that contemporaneous Brent price changes transmit weakly to inflation and industrial production.
2. **Adverse post-crisis effects in emerging importers.** The post-crisis dummy captures higher inflation and a $\sim 3.8\%$ decline in industrial production, reflecting the structural impact of the Gulf Crisis on energy-dependent economies.
3. **Asymmetric insulation in advanced exporters.** These economies display a stronger post-crisis CPI response and a modest increase in industrial production, consistent with the energy-revenue channel offsetting adverse effects.

IV. Interpretation

The results suggest that the macroeconomic effects of the 2026 Gulf Crisis were transmitted primarily through **exchange rates**, **global uncertainty**, and **post-crisis structural changes** rather than through a direct oil-price channel.

In the emerging oil-importing economies, oil shocks were not statistically significant in either the CPI or IPI equations, indicating limited direct pass-through of Brent price changes. However, exchange-rate movements and global uncertainty significantly affected both inflation and industrial production. The post-crisis period was associated with higher inflation and lower industrial production, confirming the adverse macroeconomic effects of the Gulf Crisis on energy-dependent economies.

In contrast, advanced oil-exporting economies appeared more resilient: industrial production *increased* after the crisis and inflationary pressures remained manageable. These contrasting outcomes suggest that the economic impact of geopolitical shocks depends not only on oil prices themselves, but also on countries' energy positions and macroeconomic

structures.

V. Conclusion

This study analysed the impact of the 2026 Gulf Crisis on inflation and industrial production using panel data models for emerging oil-importing and advanced oil-exporting economies. The results indicate that the direct effect of oil price shocks was generally insignificant, while exchange rates, global uncertainty, and the post-crisis structural environment played a more important role in explaining macroeconomic outcomes.

Emerging importers experienced higher inflation and weaker industrial performance following the crisis, whereas advanced exporters showed greater resilience and even modest gains in industrial activity. Overall, the findings support the view that **vulnerability to geopolitical energy shocks depends largely on a country's dependence on imported energy and its ability to absorb external disturbances.**

These results highlight the importance of:

- **Energy diversification** to reduce exposure to commodity price volatility;
- **Exchange-rate stability** to contain imported inflation;
- **Effective policy responses** during periods of geopolitical uncertainty.